

# Ahsan Bilal

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## RESEARCH INTERESTS

**Deep Learning Optimization and LLM Inference:** Optimization theory for deep learning and reasoning-oriented LLMs, including post-training methods, preference optimization, alignment, and test-time compute scaling for autoregressive and diffusion language models. Analytical foundations span statistics, probabilistic modelling, and algorithms for scalable NLP and ML systems.

**Agentic and Adaptive AI Systems:** Design of evolving multi-agent systems with adaptive behaviors, dynamic reputation modelling, and social graph-based coordination. Focus on building scalable, distributed system architectures with strong compliance and connectivity technologies for cross-functional AI product deployment.

**Reinforcement Learning for Reasoning and Decision Systems:** RL methods for reasoning-oriented LLMs, process reward models (PRMs), and agentic test-time compute allocation, with applications in complex non-stationary environments, signal processing, and ML-driven wireless communication systems (including 5G).

## EDUCATION

**University of Oklahoma**, Oklahoma, USA 2024–Present  
*M.S. in Computer Science* (Deep Learning, Reinforcement Learning, & LLMs)

- CGPA: 3.86/4.0
- Advisor: [Dr. Dean Hougén](#): Research on deep learning optimization, robust LLM reasoning, and RL-based agentic systems.

**National University of Sciences & Technology (NUST-SEECS)**, Islamabad, Pakistan Aug. 2020 – June 2024

*Bachelor of Engineering in Electrical Engineering*

- CGPA: 3.66/4.0 | Specialization GPA: 4.0/4.0 | Merit Scholarship 2020
- Senior Design Thesis: Person identification using gait with fused graph and 3D-convolutional architectures ([Presentation](#))

## RESEARCH EXPERIENCE

**REAL Lab, University of Oklahoma** Graduate Researcher  
Advisor: [Dr. Dean Hougén](#) Aug 2024 – Present

- Research on adaptive test-time compute for LLMs, including verifier/PRM-guided reasoning, agentic RL controllers, trajectory-level search, and dynamic compute allocation across autoregressive (ICML'26) and diffusion language models (COLM'26). Written technical specifications and rigorous statistical analysis for each system; communicated findings cross-functionally across collaborating institutions following collaborative teamwork best practices.
- Work on structured generative models (discrete diffusion), test-time scaling during denoising, and continuous learning diffusion models for wireless channel prediction under distribution shift (ICASSP'26). Maintained well-documented, modular codebases; produced written technical documentation for all research deliverables.

**Stanford AI Lab (SAIL), Stanford University** Research Collaborator  
Supervisors: [Prof. John M. Cioffi](#), [Dr. Emily Fox](#) | Collaborator: [M. A. Mohsin](#) Aug 2024 – Present

- Developed diffusion-based wireless estimation and neural Gaussian radio fields (nGRF), including verifier-guided trajectory search for diffusion test-time scaling and adaptive compute allocation. Applied signal processing techniques and

machine learning (ML) pipelines using pandas and scikit-learn for data preprocessing, classification tasks, and large-scale analysis.

- Worked on continuous learning under distribution shift and multimodal RAG pipelines for wireless systems. Collaborated with stakeholders from Stanford Statistics, Glasgow, and NUST in a cross-functional, collaborative research environment; publications in NeurIPS'25, ICML'25, AAAI'25, ICC'25 (Best Paper), KDD'26 (under review).

#### Conoid.ai (Internet of Agents Platform)

Research / Engineering

Ongoing

- Building a decentralized Internet of Agents platform for self-evolving AI systems, enabling dynamic agent teams based on trust, memory, and prior performance. Designing scalable distributed system architecture and HTTP-based agent communication protocols with security built into all transactions; written system specifications covering all key use cases and acting as a self-starter driving product development across the full engineering flow.

#### Undergraduate Researcher

Optimal ML Lab, NUST

Supervisor: [Dr. Ahmad Salman](#)

Jan 2023 – June 2024

- Built robust CV/biometric models: boosted-attention ViT for shadow removal and fused GCN + 3D-CNN for gait recognition. Applied analytical methods with attention to detail to achieve state-of-the-art performance on benchmark datasets; supervised experimental pipelines from problem formulation through evaluation.
- Papers under review: Shadow Removal with Boosted Attention, Gait ID using Fused Graph + 3D-CNN.

## INDUSTRY EXPERIENCE

#### Machine Learning Engineer

Islamabad, Pakistan

[Cowlar Design Studio](#) (*Y Combinator 21*) — Based in USA

Feb 2024 – Aug 2024

- Developed Action Recognition system for Smart Carts with 95% accuracy; built dual inference deployment for edge devices and Nvidia GPU clusters for scalable, real-time inference. Implemented clean, modular C++ and Python scripting components for high-throughput data processing pipelines, leveraging rapid prototyping to meet fast-paced development cycles.
- Automated fiber cable alignment using computer vision and machine learning with 96% success, improving precision to 5 micrometers and scaling production 40x. Communicated technical status and analytical findings clearly to cross-functional teams and key stakeholders.
- Contributed to AI-powered retail solutions deployed in collaboration with Al Meera Consumer Goods Company (Qatar), a ~QAR 2.9B+ annual revenue company with ~QAR 2.7B market capitalization. Delivered innovative automation addressing a core business problem; worked collaboratively with product and infrastructure teams, demonstrating strong teamwork across all phases of the project. Implemented security controls for high-volume retail transactions and used Dash-based dashboards for internal monitoring and analysis.

## SELECTED PUBLICATIONS

#### [Google Scholar](#)

[PC1] What If We Allocate Test-Time Compute Adaptively? [ICML'26](#)

A. Bilal\*, A. Mohsin\*, M. Umer, A. Subhan, H. Rizwan, A. Mohsin, D.F. Hougen

[PC2] Distilling Disagreement at Test Time Reasoning  
(Submitted)

[NeurIPS'26](#)

A. Bilal\*, M.A. Mohsin\*, M. Umer, E. Fox, D.F. Hougen

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\* indicates equal contribution.

[PC3] **General Preference Reinforcement Learning** [NeurIPS'26 \(Submitted\)](#)  
M. Umer, M.A. Mohsin, A. Bilal, E. Fox

[PC4]  **$S^3$ : Stratified Scaling Search for Test-Time in Diffusion Language Models** [COLM'26 \(Submitted\)](#)  
A. Bilal\*, M.A. Mohsin\*, M. Umer, D.F. Hougen

[PC5] **Pressure, What Pressure? Sycophancy Disentanglement in Language Models via Reward Decomposition** [COLM'26 \(Submitted\)](#)  
M.A. Mohsin\*, A. Bilal\*, M. Umer, E. Fox

[PC6] **Continuous-Utility Direct Preference Optimization** [EMNLP'26 \(In Progress\)](#)  
M.A. Mohsin\*, M. Umer\*, A. Bilal, Z. He, M.U. Rafique, A. Aali, M.A. Jamshed, J.M. Cioffi, E. Fox

[PC7] **ITDPDM: Information-Theoretic Discrete Poisson Diffusion Model** [NeurIPS'25](#)  
S. Bhattacharya, A.R. Gorle, A. Bilal, C. Ding, A.K.S. Yadav, T. Weissman

[PC8] **On the Fundamental Limits of LLMs at Scale** [TMLR'26 \(Submitted\)](#)  
A. Mohsin, A. Bilal, W. Zhao, M. Umer, Researchers from DeepMind and Meta

[PC9] **Neural Gaussian Radio Fields for Channel Estimation** [KDD'26 \(Submitted\)](#)  
M. Umer\*, M.A. Mohsin\*, A. Bilal\*, J.M. Cioffi

[PC10] **Channel Prediction Under Network Distribution Shift Using Continual Learning-Based Loss Regularization** [ICASSP'26](#)  
M.A. Mohsin, M. Umer, A. Bilal, M.I. Qadir, M.A. Jamshed, D.F. Hougen, J.M. Cioffi

[PC11] **Conditional Prior-Based Non-Stationary Channel Estimation Using Accelerated Diffusion Models** [ICASSP'26](#)  
M.A. Mohsin\*, A. Bilal\*, M. Umer, A. Ali, M.A. Jamshed, D.F. Hougen, J.M. Cioffi

[PC12] **Transformer-Based Sparse CSI Estimation for Non Stationary Channels** [ICC'26](#)  
M.A. Mohsin, M. Umer, A. Bilal, H. Rizwan, S. Bhattacharya, M.A. Jamshed, J.M. Cioffi

[PC13] **Continual Learning for Wireless Channel Prediction** [ICML'25](#)  
M.A. Mohsin, M. Umer, A. Bilal, M.A. Jamshed, J.M. Cioffi

[PC14] **Task Aware Distributed Source Coding for Correlated Audio Signals Using Perceptual Loss** [AAAI'25](#)  
S. Bhattacharya, M.A. Mohsin, A. Bilal, J.M. Cioffi

[PC15] **Retrieval Augmented Generation with Multi-Modal LLM Framework for Wireless Environments** [ICC'25](#)  
M.A. Mohsin, A. Bilal, S. Bhattacharya, J.M. Cioffi

[PC16] **HDRL for Spectrum Resource Optimization in Integrated Terrestrial and Non-Terrestrial Networks** [AAAI'25](#)  
M.A. Mohsin, H. Rizwan, M. Umer, S. Bhattacharya, A. Bilal, J.M. Cioffi

[PC17] **Abstract – LLM for Explainable AI** [IEEE DSAA'24](#)  
A. Bilal, B. Lin

[PC18] **Meta-Thinking in LLMs via Multi-Agent Reinforcement Learning: A Survey** [IEEE TAI \(Submitted\)](#)

A. Bilal, M.A. Mohsin, M. Umer, M.A.K. Bangash, M.A. Jamshed

[PC19] **On Shadow Removal With Boosted Attention in a Vision Transformer** [Springer ML \(Submitted\)](#)

A. Bilal, A. Salman, K. Khurshid, D.F. Hougen

[PC20] **Person Identification using Gait with Fused Graph and 3D Convolutional Architectures** [ACM TAIS \(Submitted\)](#)

A. Bilal, A. Salman, K. Khurshid

## TEACHING EXPERIENCE

**Teaching Assistant**

University of Oklahoma

[CS-1313: Programming for Non-majors in C](#)

Fall 2024 – Present

- Supervised weekly lab sessions; designed assignments, graded with clear rubrics, and led help sessions supporting students with C programming. Collaborated with Dr. Neeman to develop supplementary written technical documentation; communicated clearly with students in a collaborative classroom environment to address individual problem areas and foster analytical thinking.

## REVIEWER AND TALKS

**Conference Reviewer:** ICML, NeurIPS, ICASSP, PAKDD.

**Journal Reviewer:** TMLR, IEEE WCM, Springer MT&A, IP&M, Aquaculture Int., IJIM, IEEE Access, ICES, ISFI.

**Talks:** Gave a talk on “AI in Healthcare” at Norman Regional Hospital under [Dr. Lubna Mirza](#).

## HONORS AND ACHIEVEMENTS

**Honorary Certificate of Appreciation:** [IEEE Communications Society Student Competition 2025](#) for “Democratizing 6G: AI-Native Wireless Digital Twin for Global Digital Equity and Sustainability.”

**Best Student Presentation Runner-up Award:** [IEEE DSAA’24](#) Student Forum.

**Graduate Fellowship:** Awarded Gallogly College of Engineering Graduate Fellowship 2025.

**Best Paper Award:** [ICC Workshop 2025](#) in Montreal.

**Student Travel Grant:** [IEEE DSAA 2024](#) in San Diego.

**Best Adjudged Industrial Project Award:** Final Year Project received [1st place at NUST Open House 2024](#).

**UGRIP Selection:** Selected as Undergraduate Research Intern for First Cohort by MBZUAI.

**IEEE Recognition:** Selected as an Emerging Young Researcher in the IEEE Islamabad Section.

**Prime Minister Laptop Scheme:** Winner of scholarship program.

## CERTIFICATIONS

Google UX Professional Certificate

Deep Learning Specialization Certificate

The Advanced Communication Skills Course

## SKILLS

**Programming:** Python, C/C++, Java, Embedded C, MATLAB. Proficient in scripting, algorithm design, and data structures.

**Deep Learning & ML:** PyTorch, TensorFlow (Keras), OpenCV, Docker, Mlflow, EC2 Instance. Strong machine learning background spanning neural network architectures, natural language processing (NLP), continuous learning, and scalable pipelines for large-scale data processing and dataset management. Proficient in Dash for interactive data analysis and visualization.

**Statistics & Theory:** Analytical and statistical learning theory, probabilistic modelling, information theory, and optimization methods applied to machine learning research; experienced in rigorous analysis and problem formulation from specification through evaluation.

**Infrastructure & Systems:** Distributed systems design, HTTP APIs, Docker-based infrastructure with security best practices, scalable deployment on cloud and edge. Familiar with development best practices, including version control, clean code, and written technical documentation; applied rapid prototyping to iterate quickly across diverse use cases.

**Web Development:** JavaScript (React.js), HTML/CSS, Next.js.

**Tools:** VS Code, Git, AutoCad, Figma, PyCharm, Raspberry Pi OS, NginX.

**Soft Skills:** Collaborative teamwork across cross-functional teams; communicate technical status and findings clearly to diverse audiences and stakeholders; supervised junior researchers and student groups; self-starter with attention to detail and innovative problem-solving in fast-paced environments.

**Design:** Figma, Adobe XD, Adobe Illustrator, Adobe Photoshop, Sketch, WordPress Theme Design.

**Core Competencies:** Signal Processing, Machine Learning (ML), Classification, Algorithms, 5G, Connectivity Technologies, Compliance, Data Analysis (pandas), Model Development (scikit-learn), Distributed AI Systems.